Node.js — Day 19

HTTP/API Concepts — Complete Notes

■ What we covered

HTTP headers, **Content-Type**, **Accept**, request vs response headers, and the difference between Accept and Content-Type.

1. HTTP Headers

HTTP headers are metadata (additional information) about an HTTP request or response.

Think of a header like the label on a package ■: the body is the actual data, while the header gives information about that data.

Status → what happened **Headers** → information about the data **Body** → actual data

2. Content-Type

Content-Type tells the receiver what format the body is in. It can be used in both requests and responses.

Client → Server:

```
Content-Type: application/json
```

Meaning: “The request body I'm sending is intended to be JSON.”

Server → Client:

```
Content-Type: application/json
```

Meaning: “The response body I'm sending is JSON.”

The header declares the intended format; the server can still validate whether the body is actually valid JSON.

3. Accept

Accept tells the server what response format the client prefers to receive.

```
Accept: application/json
```

Meaning: “I want the response in JSON.”

4. Accept vs Content-Type

Header	Meaning	Typical direction
Accept	What response format I want	Client → Server
Content-Type	What format the body I am sending is	Client → Server or Server → Client

Easy memory trick: Accept = “I want” ■ Content-Type = “I am sending” ■

5. Node.js Example

```
const http = require("http");

const server = http.createServer((req, res) => {
```

```
  res.writeHead(200, {
    "Content-Type": "application/json"
  });

  res.end(JSON.stringify({
    message: "Hello"
  }));
});

server.listen(3000);
```

Here, **200** is the status code, **Content-Type** is a response header, and the JSON object is the response body.

6. Checkpoint

You correctly identified that headers provide metadata and that **Accept: application/json** means the client wants a JSON response.

■ Final Recap

• HTTP headers provide metadata about requests/responses.
• Content-Type describes the format of the body being sent.
• Content-Type can appear on both requests and responses.
• Accept describes the response format the client wants.
• Accept is commonly sent by the client in a request.
• Accept = I want; Content-Type = I am sending.

■ Progress

Day 19: COMPLETE ■
Next planned topic: Better Error Handling.